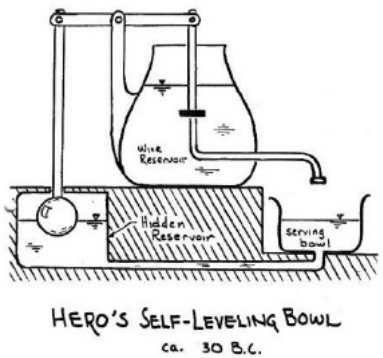
What is a control system?

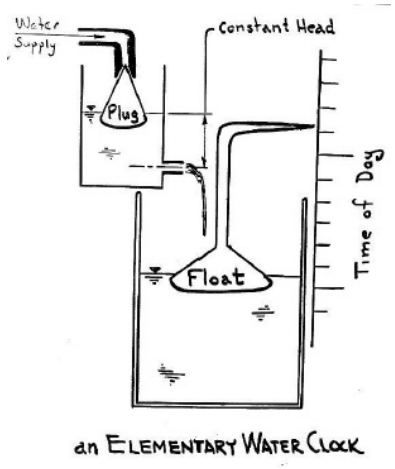
- Generally speaking, a control system is a system that is used to realize a desired output or objective.
- Control systems are everywhere
  - They appear in our homes, in cars, in industry, in scientific labs, in hospitals ...
  - Principles of control have an impact on diverse fields as engineering, aeronautics, economics, biology and medicine ...
  - Wide applicability of control has many advantages (e.g., it is a good vehicle for technology transfer)

A brief history of control

- Two of the earliest examples
  - Water clock (270 BC)
  - Self-leveling wine vessel (100BC)



(a) Self-leveling wine vessel

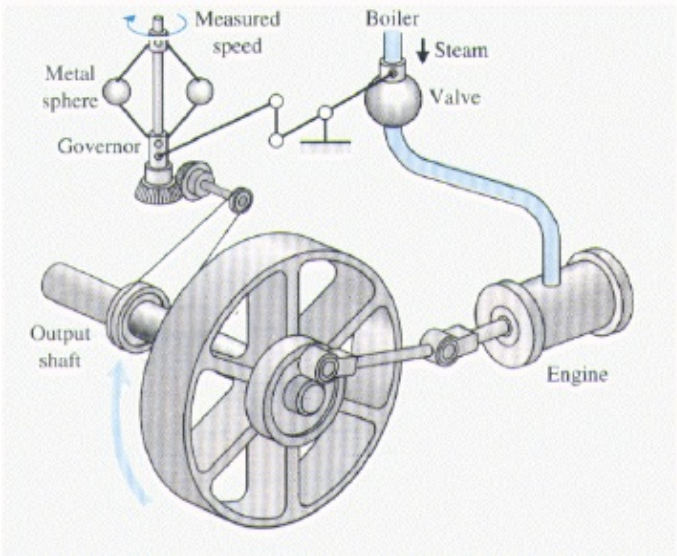


(b) Water clock

Figure: Earliest examples of control systems

A brief history of control

- Fly-ball governor (James Watt, 1769)
  https://www.youtube.com/watch?v=dALXNAIeXd4





- the first modern controller
- regulated speed of steam engine
- reduced effects of variances in load
- propelled Industrial Revolution

A brief history of control

Birth of mathematical control theory

- G. B. Airy (1840)
  - the first one to discuss instability in a feedback control system
  - the first to analyze such a system using differential equations
- J. C. Maxwell (1868)
  - the first systematic study of the stability of feedback control
- E. J. Routh (1877)
  - deriving stability criterion for linear systems
- A. M. Lyapunov (1892)
  - deriving stability criterion that can be applied to both linear and nonlinear differential equations
  - results not introduced in control literature until about 1958

A brief history of control

Birth of classical control design method

- H. Nyquist (1932)
  - developed a relatively simple procedure to determine stability from a graphical plot of the loop-frequency response.
- H. W. Bode (1945)
  - frequency-response method
- W. R. Evans (1948)
  - root-locus method

With the above methods, we can design control systems that are stable, acceptable but not necessarily optimal in some pre-specified sense.

A brief history of control

Development of modern control design

- Late 1950s: designing optimal systems in some meaningful sense
- 1960s: digital computers help time-domain analysis of complex systems, modern control theory has been developed to cope with the increased complexity of modern plants
- 1960s-1980s: optimal control of both deterministic and stochastic systems; adaptive control and learning control
- 1980s-present: robust control, H-inf control ...

A brief history of control

Recent applications of modern control theory include such non-engineering systems as biological, biomedical, economic and socioeconomic systems ...

Basic components of a control system

- Plant
- Controlled variable
- Expected value
- Controller
- Actuator
- Sensor
- Disturbance

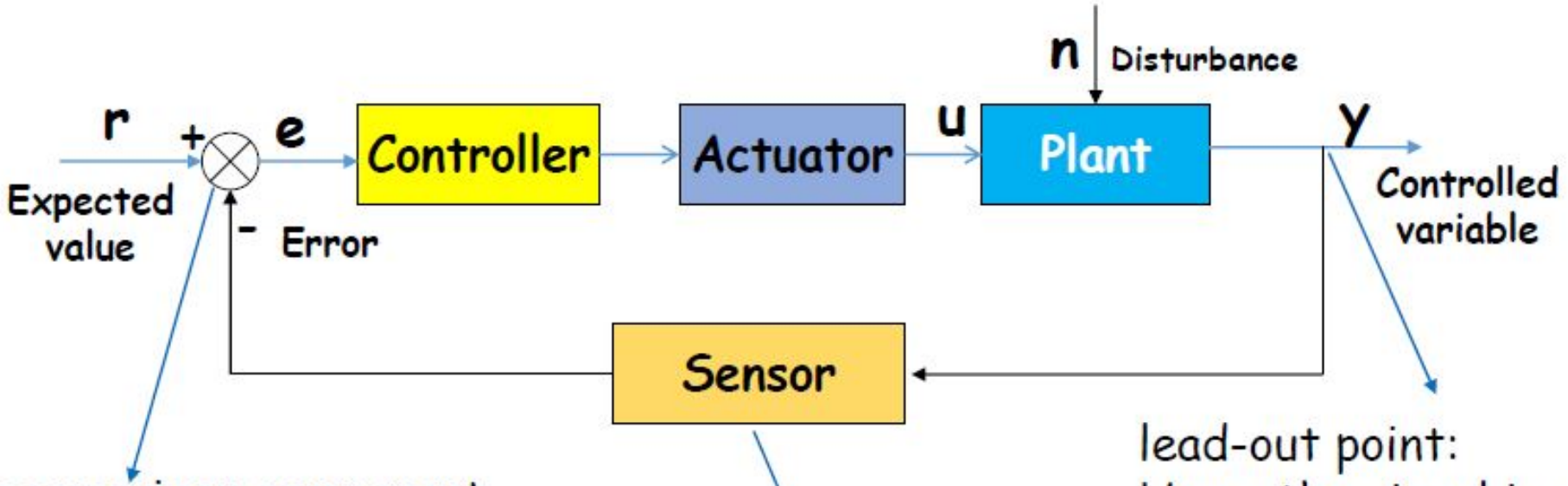
Basic components of a control system

1. **Plant:** a physical object to be controlled such as a mechanical device, a heating furnace, a chemical reactor or a spacecraft.
2. **Controlled variable:** the variable controlled by Automatic Control System, generally refers to the system output.
3. **Expected value:** the desired value of controlled variable based on requirement, often it is used as the reference input.

Basic components of a control system

4. **Controller:** an agent that can calculate the required control signal.
5. **Actuator:** a mechanical device that takes energy, usually created by air, electricity, or liquid, and converts that into form suitable for the plant.
6. **Sensor:** a device that measures a physical quantity and converts it into a signal which can be read by an observer or by an instrument.
7. **Disturbance:** the unexpected factors disturbing the normal functional relationship between the controlling and controlled parameter variations.

Block diagram of a control system



comparison component (comparison point) : its output equals the algebraic sum of all input signals.

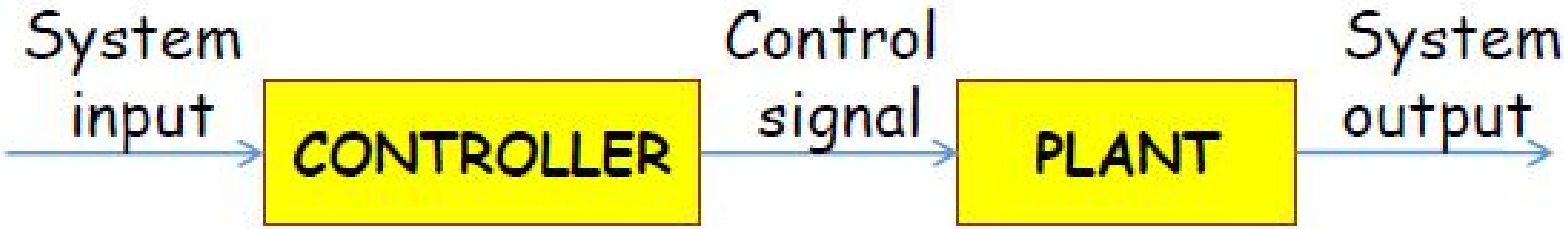
"+": plus; "-": minus

The Block represents the function and name of its corresponding mode, we don't need to draw detailed structure, and the line guides for the transfer route.

lead-out point: Here, the signal is transferred along two separate routes.

Open loop Control Systems

- Open loop control systems: those systems in which the output has no effect on the control action.



- The output is neither measured nor fed back for comparison with the input.
- For each reference input, there corresponds a fixed operating conditions; the accuracy of the system depends on calibration.
- In the presence of disturbances, an open-loop system will not perform the desired task.

Open loop Control Systems

Examples:

- Washing machine



- Traffic signals



Note that any control system that operates on a time basis is open loop.

Open loop Control Systems

Some comments on open loop control systems

+ Simple construction and ease of maintenance.
+ Less expensive than a closed-loop system.
– Recalibration is necessary from time to time.
– Sensitive to disturbances and uncertainties, so less accurate.

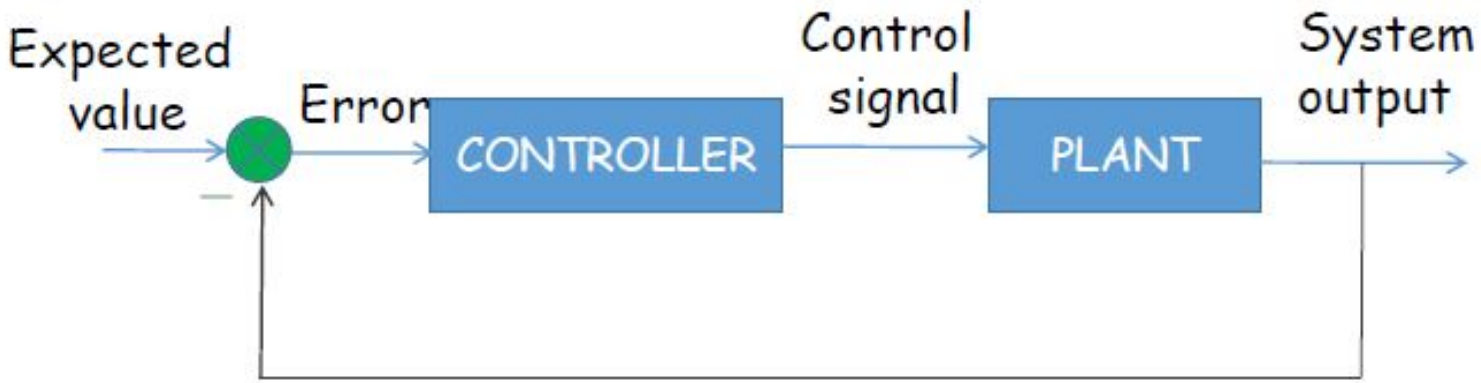
Open loop Control Systems

When should we apply open loop control?

- The relationship between the input and output is exactly known.
- There are neither internal nor external disturbances.
- Measuring the output precisely is very hard or economically infeasible.

Closed loop control systems

- Closed-loop control systems are often referred to as feedback control systems.
- The idea of feedback:
  - Compare the actual output with the expected value.
  - Take actions based on the difference (error).



  - This seemingly simple idea is tremendously powerful.
  - Feedback is a key idea in the discipline of control.

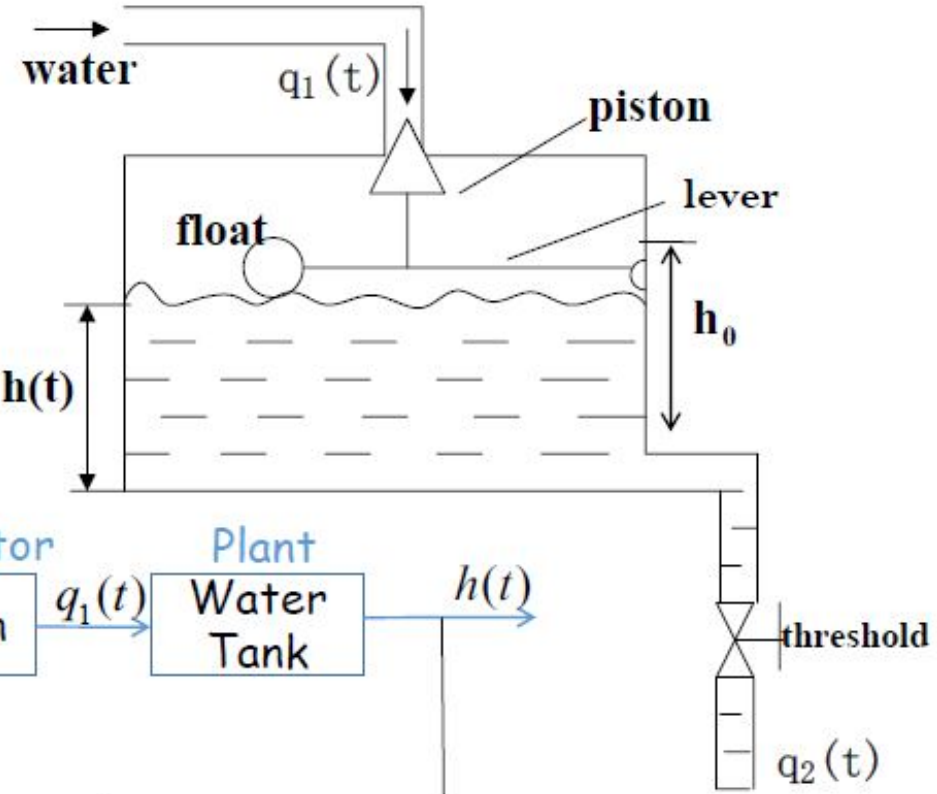
Closed loop control systems

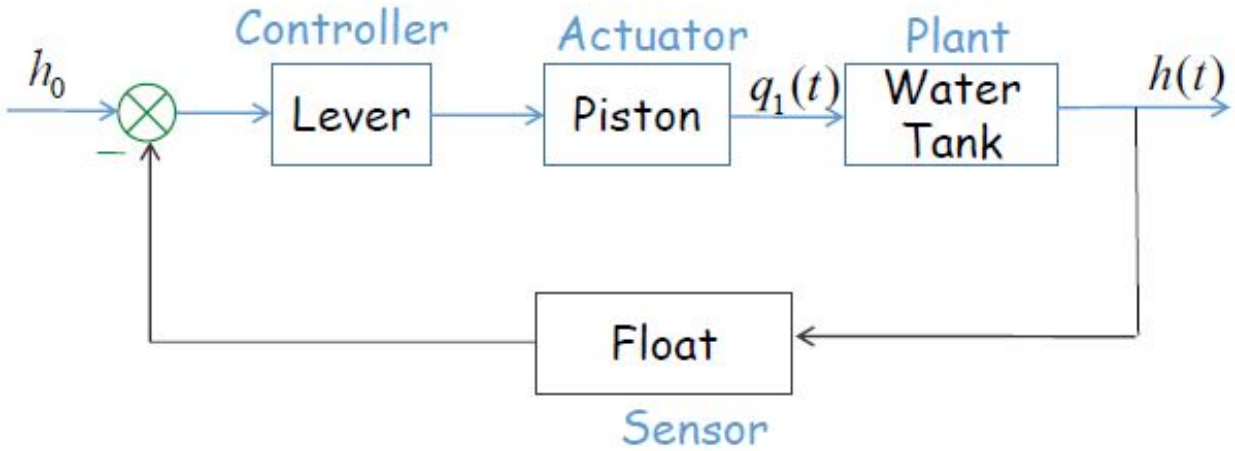
- In practice, feedback control system and closed-loop control system are used interchangeably.
- Closed-loop control always implies the use of feedback control action in order to reduce system error.

Example 1: Flush toilet

https://www.youtube.com/watch?v=yEBMUDL0apE

Plant: water tank
Input: water flow
Output: water level $h(t)$
Expected value: h_0
Sensor: float
Controller: lever
Actuator: piston

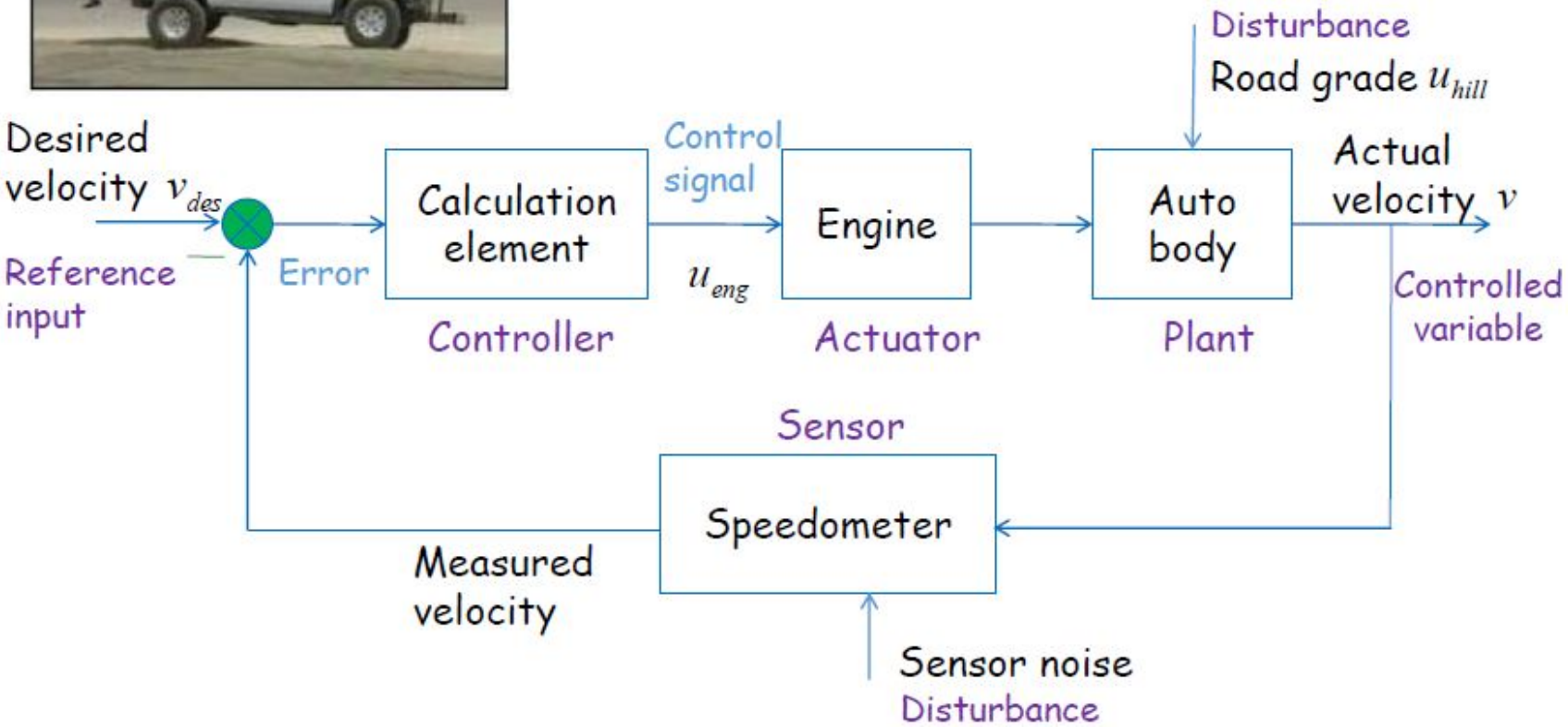




Example 2: Cruise control



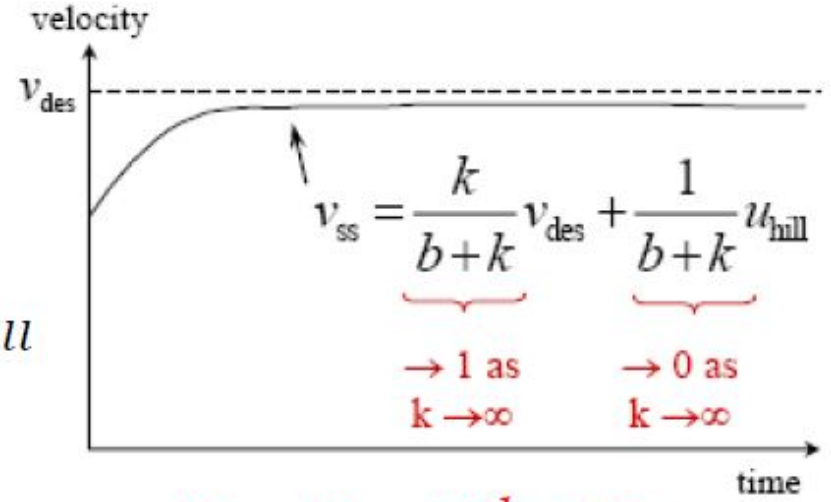
$$m\dot{v} = -bv + u_{eng} + u_{hill}$$
$$u_{eng} = k(v_{des} - v)$$



Example 2: Cruise control



$$m\dot{v} = -bv + u_{eng} + u_{hill}$$
$$u_{eng} = k(v_{des} - v)$$



$$v_{ss} \to v_{des} \text{ as } k \to \infty$$

- Stability/performance
  - Steady state velocity approaches desired velocity as k $to\infty$
  - Smooth response: no overshoot or oscillations
- Disturbance rejection
  - Effect of disturbances (e.g., hills) approaches zero as $k \to \infty$
- Robustness
  - Results do not depend on the specific values of b, m or k, for k sufficiently large.

Note

- In this example, we ignore the dynamic response of the car and consider only the steady behavior.
  - Dynamics will play a major role in later chapters.
- There are limits on how high the gain k can be made.
  - when dynamics are introduced, the feedback can make the response worse than before, or even cause the system to be unstable.

Comments on feedback control

Main advantages of feedback:

- reduce disturbance effects
- make system insensitive to variations
- stabilize an unstable system
- create well-defined relationship between output and reference

Comments on feedback control

Potential drawbacks of feedback:

- may cause instability if not used properly
- couple noise from sensors into the dynamics of a system
- increase the overall complexity of a system

Comments on feedback control

Feedback control design:

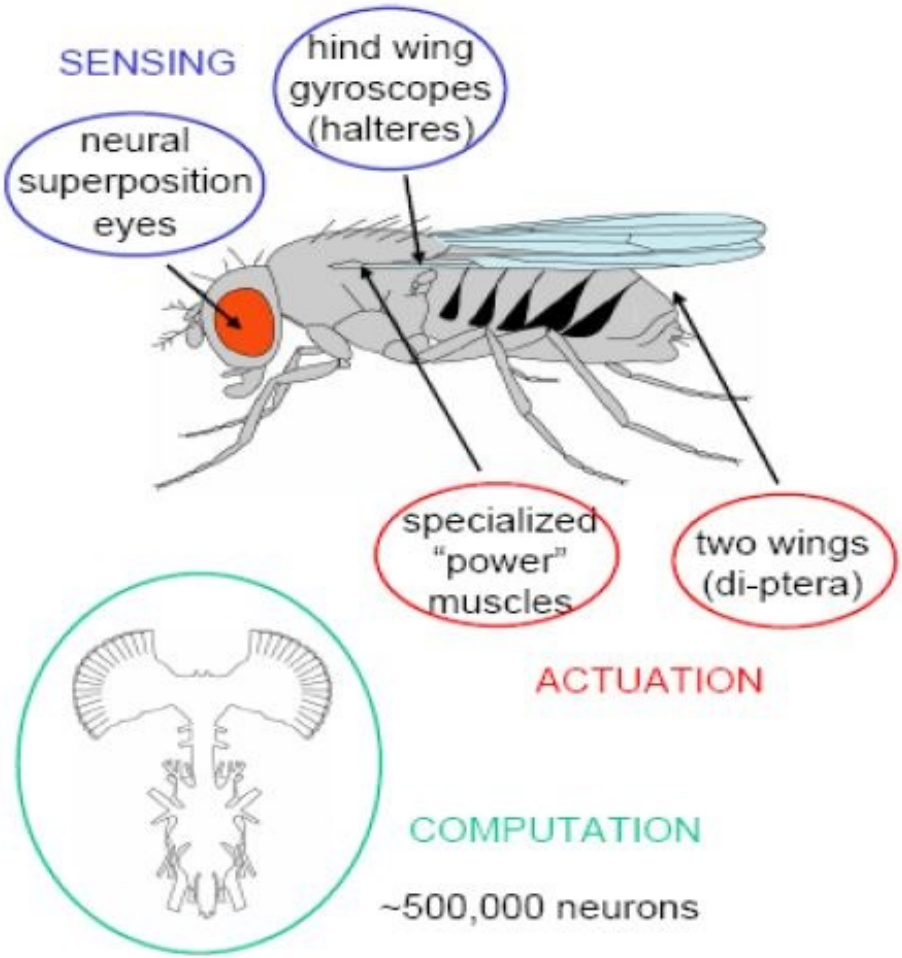
- how to **get the gain as large as possible to reduce the error**

  without

  making the system become unstable

Other examples of feedback

- Feedback systems are not limited to engineering but can be found in various non-engineering fields as well.



Other examples of feedback

- The human body is highly advanced feedback control system.
- Body temperature and blood pressure are kept constant by means of physiological feedback.
- Feedback makes the human body relatively insensitive to external disturbance. Thus we can survive in a changing environment.

Open-loop vs. closed-loop

- Open loop control
  - Simple structure, low cost
  - Easy to regulate
  - Low accuracy and resistance to disturbance
- Closed loop control
  - Ability to correct error
  - High accuracy and resistance to disturbance
  - Complex structure, high cost
  - Selecting parameter is critical (may cause stability problem)

Open loop + Closed loop = Composite control system

Thinking time

Think of some examples of open-loop control and closed-loop control systems?

For each system, could you identify the sensor, actuator and controller?

Classification of control systems

- According to structure
  - Open loop control
  - Closed loop control
  - Composite control

Classification of control systems

- According to reference input
  - Constant-value control
    - the reference input (expected value) is a constant value
    - the controller works to keep the output around the constant value
    - Examples: constant-temperature control, liquid level control and constant-pressure control.
  - Servo/tracking control
    - the reference input may be unknown or varying
    - the controller works to make the output track the varying reference
    - Examples: automatic navigation systems on boats and planes, satellite-tracking antennas
  - Programming control
    - the input changes according to a program
    - the controller works according to predefined command
    - Examples: numerical control machine

Classification of control systems

- According to system characteristics
  - Linear control systems
    - superposition principle applies

$$f(x_1) = y_1 \quad f(x_2) = y_2$$

$$\Downarrow$$

$$f(\alpha_1 x_1 + \alpha_2 x_2) = \alpha_1 y_1 + \alpha_2 y_2$$

    - described by linear differential equation
  - Nonlinear control systems
    - described by nonlinear
    - differential equation

Remarks on nonlinear systems

- Quite often, nonlinear characteristics are intentionally introduced in a control system to improve its performance or provide more effective control.
- For instance, to achieve minimum-time control, an on-off (bang-bang or relay) type controller is used in many missile or spacecraft control systems
- There are no general methods for solving a wide class of nonlinear systems

Classification of control systems

- According to signal form
  - Continuous time control system
    - All the signals are functions of continuous time variable t
  - Discrete time control system
    - Signals are in the form of either a pulse train or a digital code

Remarks on digital control systems

- A digital control system refers to the use of a digital computer or controller in the system, so that the signals are digitally coded, such as in binary code.
- Digital computers provide many advantages in size and flexibility.
  - The expensive equipment used in a system may be shared simultaneously among several control channels.
  - Digital control systems are usually less sensitive to noise.

Basic requirements for control systems

- **Stability:** refers to the ability of a system to recover equilibrium
- **Quickness:** refers to the duration of transient process before the control system reaches its equilibrium
- **Accuracy:** refers to the size of steady-state error when the transient process ends

Steady-state error = desired output – actual output

Note

- For a control system, the above three performance indices (stability, quickness, accuracy) are sometimes contradictory.
- In design of a practical control system, we always need to make compromise.

Review questions

True or false?

1. A closed-loop control system is usually more accurate than an open-loop system. (T/F)
2. Feedback is sometimes used to improve the sensitivity of a control system. (T/F)
3. If an open-loop system is unstable, then applying feedback will always improve its stability. (T/F)
4. Feedback can cause instability. (T/F)
5. Nonlinear elements are sometimes intentionally introduced to a control system to improve its performance. (T/F)
6. Discrete-data control systems are more susceptible to noise due to the nature of its signals. (T/F)